Faculty of Engineering and Technology

Electrical and Computer Engineering Department

OPERATING SYSTEMS

ENCS3390

Programming Task 2

Prepared by:

Raghad Jamhour 1220212

Yasmin Al Shawawrh 1220848

Date: 12/01/2025

Some Test Cases:

```
1   0   1   CPU {20}    IO{30}     CPU {40}
0   1   1   CPU {R[1], R[2], 50, F[2], F[1]}
```

```
C:\Users\HP\PycharmProjects\osproject\.venv\Scripts\python.exe C:\Users\HP\PycharmProjects\osproject\main.py
CPU burst started at :0 for 1
CPU burst ended at :20 for 1
PID 1 performs IO burst for 30 units (time: 20 to 50)
CPU burst started at :20 for 0
CPU burst ended at :70 for 0
CPU burst started at :70 for 1
CPU burst ended at :110 for 1
Total Waiting Time: 39
Total Turnaround Time: 179
Average Waiting Time: 19.5
Average Turnaround Time: 89.5
```

```
0   0   1   CPU {R[1], 15, R[2], 10, F[1], F[2]}
1   0   1   CPU {5, R[2], R[1], 10, F[2], F[1]}
2   0   0   CPU {50}
```

```
C:\Users\HP\PycharmProjects\osproject\.venv\Scripts\python.exe C:\Use
CPU burst started at :0 for 2
CPU burst ended at :50 for 2
PID 0 executes CPU burst for 10 units (time: 50 to 60)
PID 1 executes CPU burst for 5 units (time: 60 to 65)
PID 0 executes CPU burst for 5 units (time: 65 to 70)
Deadlock detected on resource R[1]. 1 processes are waiting.
Handling deadlock...
Aborting process PID 1 to resolve deadlock.
Aborted process PID 1 arrival time updated to 10
PID 0 executes CPU burst for 10 units (time: 70 to 80)
PID 1 executes CPU burst for 10 units (time: 80 to 90)
```

```
1    1   0   1   CPU {20}    IO{30}     CPU {40}
2    0   0   1   CPU {R[1], R[2], 50, F[2], F[1]}
```

main

```
C:\Users\HP\PycharmProjects\osproject\.venv\Scripts\python.exe C:\Use
PID 1 executes CPU burst for 10 units (time: 0 to 10)
PID 0 executes CPU burst for 10 units (time: 10 to 20)
PID 1 executes CPU burst for 10 units (time: 20 to 30)
PID 0 executes CPU burst for 10 units (time: 30 to 40)
PID 1 performs IO burst for 30 units (time: 40 to 70)
PID 0 executes CPU burst for 10 units (time: 40 to 50)
PID 0 executes CPU burst for 10 units (time: 50 to 60)
PID 0 executes CPU burst for 10 units (time: 60 to 70)
PID 1 executes CPU burst for 10 units (time: 70 to 80)
PID 1 executes CPU burst for 10 units (time: 80 to 90)
PID 1 executes CPU burst for 10 units (time: 90 to 100)
PID 1 executes CPU burst for 10 units (time: 100 to 110)
```

```
1    1   0   1   CPU {20}    IO{30}     CPU {40}
2    0   0   1   CPU {R[1], R[2], 50, F[2], F[1]}
```

main

```
C:\Users\HP\PycharmProjects\osproject\.venv\Scripts\python.exe C:\Use
PID 1 executes CPU burst for 10 units (time: 0 to 10)
PID 0 executes CPU burst for 10 units (time: 10 to 20)
PID 1 executes CPU burst for 10 units (time: 20 to 30)
PID 0 executes CPU burst for 10 units (time: 30 to 40)
PID 1 performs IO burst for 30 units (time: 40 to 70)
PID 0 executes CPU burst for 10 units (time: 40 to 50)
PID 0 executes CPU burst for 10 units (time: 50 to 60)
PID 0 executes CPU burst for 10 units (time: 60 to 70)
PID 1 executes CPU burst for 10 units (time: 70 to 80)
PID 1 executes CPU burst for 10 units (time: 80 to 90)
PID 1 executes CPU burst for 10 units (time: 90 to 100)
PID 1 executes CPU burst for 10 units (time: 100 to 110)
```